

Cloudflare Radar 2024 Year in Review

A data-driven analysis of global Internet traffic, adoption trends, connectivity disruptions, and security threats observed through Cloudflare's network in 2024

Source: Cloudflare Radar | radar.cloudflare.com/year-in-review/2024

17% Global Traffic Growth and Starlink's 3.3x Surge

17%

YoY Global Traffic Growth

3.3x

Starlink Traffic Surge

41%

Traffic from Mobile Devices

28%

IPv6 Adoption (Dual-Stack)

Normalized Starlink traffic worldwide



- ▶ Global Internet traffic grew 17% year-over-year in 2024, continuing the upward trend driven by increasing reliance on the Internet
- ▶ SpaceX Starlink traffic surged 3.3x, expanding satellite connectivity to previously underserved regions
- ▶ Starlink is not yet available globally; trend line is scaled relative to peak observed request volume
- ▶ 41% of global traffic originated from mobile devices
- ▶ IPv6 adoption reached 28% of dual-stack traffic; Spain recorded the highest download speed

Google Leads Top 10 Services as Generative AI Rises

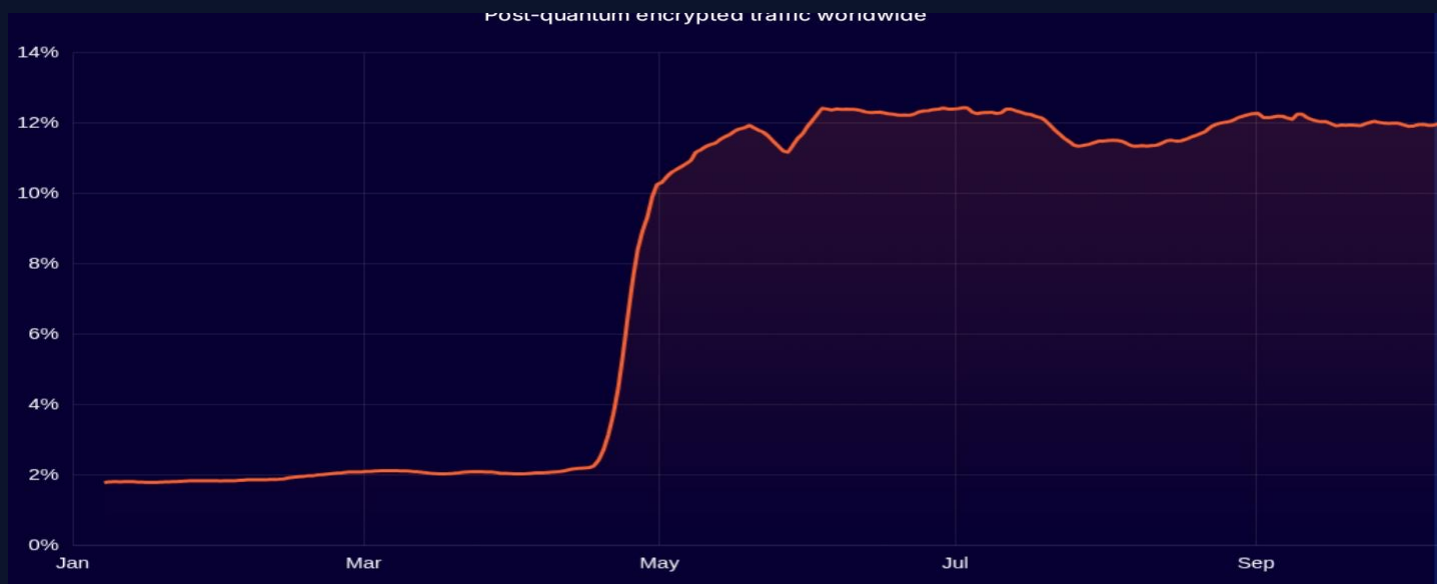
Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Generative AI: The Defining Trend of 2024

- ▶ Generative AI continued its meteoric rise throughout 2024, reshaping the Internet services landscape
- ▶ Google maintained #1 position for the second consecutive year, ranking based on Cloudflare 1.1.1.1 DNS resolver data
- ▶ The Metaverse receded into the background as generative AI dominated attention and investment

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3



2% → 13%

Post-Quantum TLS 1.3 Share

21%

HTTP/3 Traffic Adoption

Post-Quantum Encryption

- ▶ Started at ~2% of TLS 1.3 traffic in early 2024
- ▶ Chrome 124 enabled post-quantum key agreement by default on April 17, triggering rapid adoption
- ▶ Climbed to ~12-13% by year end
- ▶ Firefox rolling out post-quantum by default; Safari began initial testing

HTTP/3 & Protocol Trends

- ▶ 21% of traffic uses HTTP/3 (QUIC-based) for faster connections and built-in encryption
- ▶ 33% of traffic came from iOS devices
- ▶ Google dominates search at 89% of referral traffic

API Client Languages

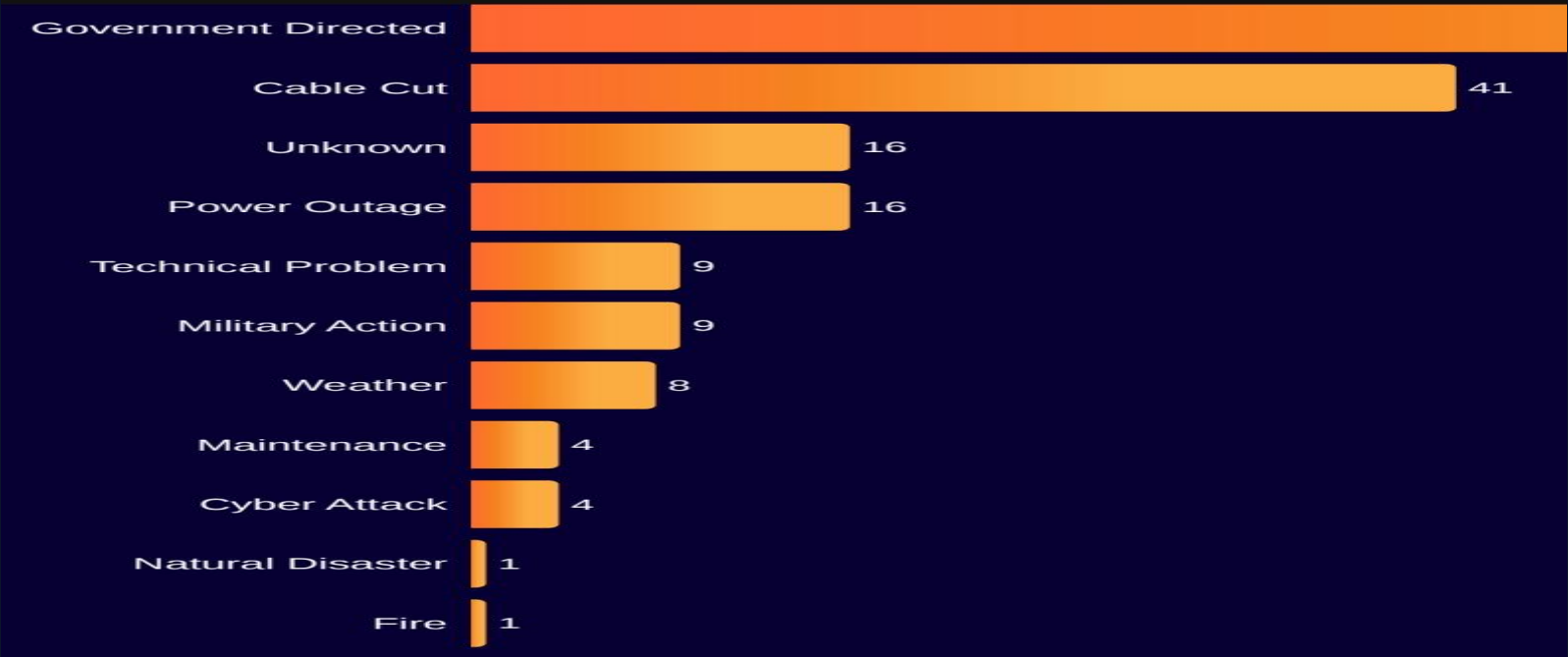
- ▶ Go is #1 API client language (~12% of automated API requests)
- ▶ Followed by Node.js, Python (9.6%), Java (7.4%), .NET (3.6%)

225 Major Disruptions: Government Shutdowns Lead Outage Causes

225

Major Internet Disruptions in 2024

Cause	Count
Government Directed	116+
Cable Cut	41
Unknown	16
Power Outage	16
Technical Problem	9
Military Action	9
Weather	8
Maintenance	4
Cyber Attack	4
Natural Disaster	1
Fire	1



Key Insight: Government-Driven Disruptions Dominate

- ▶ Government-directed shutdowns (116+) accounted for over half of all major disruptions, far exceeding any other cause
- ▶ Cable cuts were the second leading cause with 41 incidents, highlighting physical infrastructure vulnerability
- ▶ Power outages and unknown causes each contributed 16 disruptions; cyber attacks accounted for only 4

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

6.5%

Traffic Mitigated by Cloudflare

68%

Bot Traffic from Top 10 Countries

4.3%

Emails Were Malicious

6.4%

RPKI Valid Routes Increase

Threat Landscape

- ▶ 6.5% of all traffic processed by Cloudflare was mitigated, indicating the scale of ongoing threats
- ▶ Gambling and Games was the most attacked industry sector
- ▶ 68% of global bot traffic originated from just the top 10 countries
- ▶ RPKI valid routes increased by 6.4%, improving Internet routing security

Log4j: A Persistent Threat

- ▶ CVE-2021-44228 (Log4j), disclosed Dec 2021, continues to be actively targeted
- ▶ Underscores the persistent nature of widely known vulnerabilities years after disclosure

Email Security Metrics

Metric	Value
Malicious Email Rate	4.3%
Deceptive Link Share	43%
Top Malicious TLD	.bar

- ▶ 4.3% of all emails processed were classified as malicious
- ▶ 43% of malicious emails contained a deceptive link designed to trick recipients
- ▶ The .bar top-level domain originated the largest share of malicious and spam email

Five Takeaways Shaping the 2025 Internet Outlook

1

Traffic Growth Trajectory

Global Internet traffic grew 17% YoY with Starlink surging 3.3x — satellite and mobile expansion will continue driving growth into 2025, demanding scalable infrastructure investment.

2

Post-Quantum Encryption Momentum

Post-quantum encryption leapt from ~2% to 13% of TLS 1.3 traffic after Chrome 124's default enablement. With Firefox and Safari following, 2025 will be the year of mainstream PQC adoption.

3

Connectivity Fragility from Government Actions

Government-directed shutdowns caused 116+ of 225 major disruptions, far exceeding cable cuts (41) and power outages (16). Internet resilience planning must account for politically motivated outages.

4

Persistent Security Threats

6.5% of traffic was mitigated, Gambling/Games was the most targeted sector, and Log4j exploitation persists years after disclosure. Threat mitigation and vulnerability lifecycle management remain critical.

5

Evolving Developer Ecosystem

Go surpassed other languages to become the top API client language at ~12% of automated API traffic. HTTP/3 reached 21% adoption. The developer ecosystem is rapidly shifting toward modern protocols and languages.